

MASENO 2025 KCSE PREDICTION



CONFIDENTIAL EXAM

451/1

COMPUTER STUDIES

PAPER 1

TIME: 2½ HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTION TO CANDIDATES

- ❖ Write your name and index number in the spaces provided above
- ❖ *This paper consists of **two** sections **A** and **B**.*
- ❖ *Answer **ALL** questions in section **A**.*
- ❖ *Answer question **16** and any other **THREE** questions from section **B**.*
- ❖ *All answers should be written in the spaces provided on the question paper.*

For Examiners Use Only

Section	Questions	Candidates Score
A	1-15	
B	16	
	17	
	18	
	19	
	20	
	TOTAL SCORE	

1. Give **two** reasons why smaller computers like Laptops tend to be more expensive than Desktop computers.
(2 marks)
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2. a). State **two** keyboard keys that work in a state of On/Off switches in **operation** (1 mark)
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- b). State **four** factors you would consider when choosing a secondary storage device (2 marks)
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-
3. i) What is distributed data processing? (1 mark)
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- ii) Outline **two** advantages of distributed data processing in an organization (2marks)
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4. Differentiate between graphic-based and layout based desktop publishing software (2 marks)

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5. (a) Write the following acronyms in full as used in operating systems.

(i) GUI (½ mark)

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(ii) WIMP (½ mark)

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(b) State **two** factors considered when choosing an operating system for use in a computer.(2 mks)

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6. a) Define the term audit trail

(1 mark)

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b) List any **three** ways of recovering data lost through accidental erasure.

(3marks)

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7. State **three** functions of an email software.

(3marks)

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8. Name **two** responsibilities that are carried out by a computer programmer (2 marks)

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9. State **three** ways in which ICT can improve the access of News in Kenya. (3marks)

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10. Give **three** reasons why people prefer in-house developed application programs to general-purpose application packages. (3marks)

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11. State any **two** negative impacts of ICT on the environment (2 marks)

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12. Highlight **three** reasons why most network technicians prefer using command line operating systems to configure networking equipment. (3 marks)

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13. A computer teacher wants to minimize illegal entry of people to a computer laboratory. Explain **three** measures which have to be put in place in order to curb this illegal entry. (3 mks)

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14. Outline **two** benefits of mail merging. (2 marks)

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15. Without proper marketing, a business cannot survive in a competitive environment hence computers are being used in a number of ways to enhance marketing. List any **two** ways of ensuring this is effected.

(2 marks)

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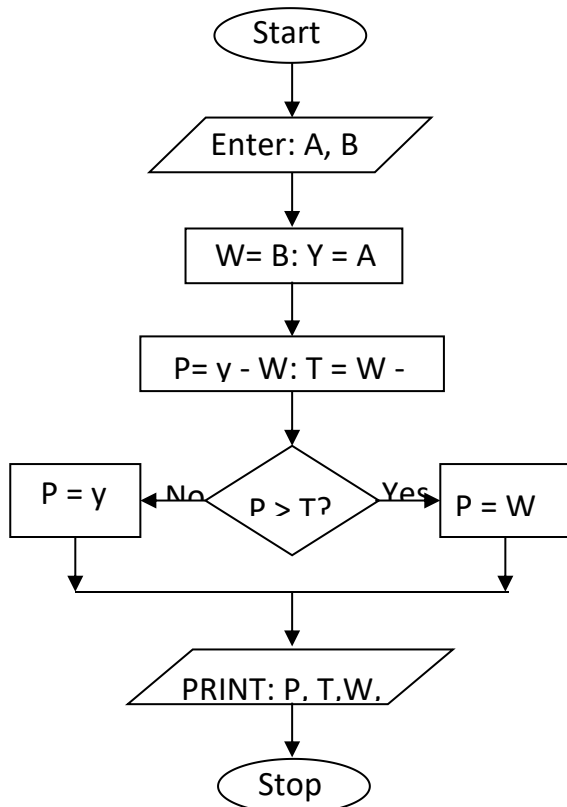
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SECTION B: (60 MARKS)

Answer question 16 (compulsory) and any other THREE questions from this section.

16. (a) Study the flowchart **below** and answer the questions that follow.



(i) State the name of the control structure shown in the flowchart. (1 mark)

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(ii) Assuming the user keys in 12 and 19 when prompted to enter the values for B and A respectively, determine the printed values of P, T, W and y. (4 marks)

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(iii) Write the pseudo code that would be used in place of the flowchart. (6 marks)

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(b) What is the difference between looping and selection? (2 marks)

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c) State any **two** advantages of low level languages.

(2 marks)

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17.

a) The following are some of the phases in the systems development life cycle (SDLC): system analysis, system design, system implementation, system review and maintenance. State **four** activities that are carried out during the system implementation phase **(4 marks)**

b) Give **three** reasons why system maintenance phase is necessary in SDLC **(3 marks)**

c) State **two** instances where observation is not a viable method of gathering information during system analysis stage **(2 marks)**

d) Various considerations should be made during input and output design. State **two** considerations for each case **(4 marks)**

e) List **two** sources of computer viruses **(2 marks)**

18. a) Use twos complement to perform the following arithmetic operations

i) $15_{10} - 12_{10}$ **(4marks)**

ii) $10111_2 - 1011_2$ **(4marks)**

b) 1011_2 is a ones complement binary representation of a number using four bits work out the likely positive equivalent in base 10. **(2marks)**

c) Convert the decimal fraction 10.375_{10} to its binary equivalent. **(4marks)**

d) Assuming the existence of base six, list the numbers that would be used in the number system.

(1mark)

19. a) i) Write the acronym DBMS in full.

(1mark)

(ii) State any **three** functions of a DBMS.

(3marks)

b) Indicate the type of cell reference depicted by the following.

(3marks)

i) H5.....

ii) H \$ 5.....

iii) \$H\$5.....

c) Mr. Maanzo is a teacher at Soyot High school and wishes to use Spreadsheet to process the marks obtained by his students during the term. The worksheet below shows the details used.

A	B	C	D	E	F	G	H	I
1	NAME	CAT-1	CAT-2	CATTOTAL	EXAM	TOTAL	GRADE	PST
2	Susan Muani	14	06		56			
3	Peter Kwena	13	08		34			
4	Mary Ann	10	04		59			
5	Paul Njeru	11	07		57			
6	Janet Frida	06	09		48			
7	Justin Gerard	08	07		50			

NB:CAT -1 and CAT -2 are out of 15 marks each and EXAM is out of 70 marks

(i) Write a function that displays “PASSED” when the total marks is >60 otherwise display “FAIL”.

(2marks)

(ii) Write a function that assigns Paul P positions according to the total marks.

(2marks)

d) Use illustrations to explain **two** text alignment features used in electronic word processors

(4 mks)

20. a) Match the following statements with repeater, router or a bridge.

(2 marks)

	STATEMENT	DEVICE
	Operates at data link layer	

	Determine the best path for data to follow	
	Regenerates signals	
	Forwards everything which it doesn't recognize	

b) Discuss two disadvantages of wireless networks. **(4 marks).**

c) Write the following abbreviations in full. **(2 marks)**

i. F.T.P

ii. H.T.T.P

d) With the aid of a diagram, discuss Hybrid topology. **(3 marks)**

e) State **two** advantages of using fiber optic cables. **(2 marks)**

(d) Explain the meaning of the following terms as used in signal transmission. **(2 marks)**

i. Attenuation

ii. Noise

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